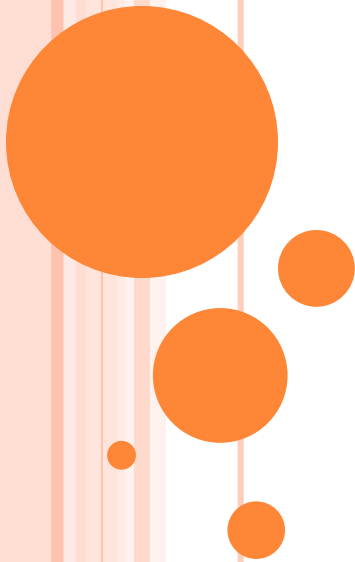


COMPUTER ORGANIZATION AND ARCHITECTURE

5. INPUT OUTPUT MANAGEMENT





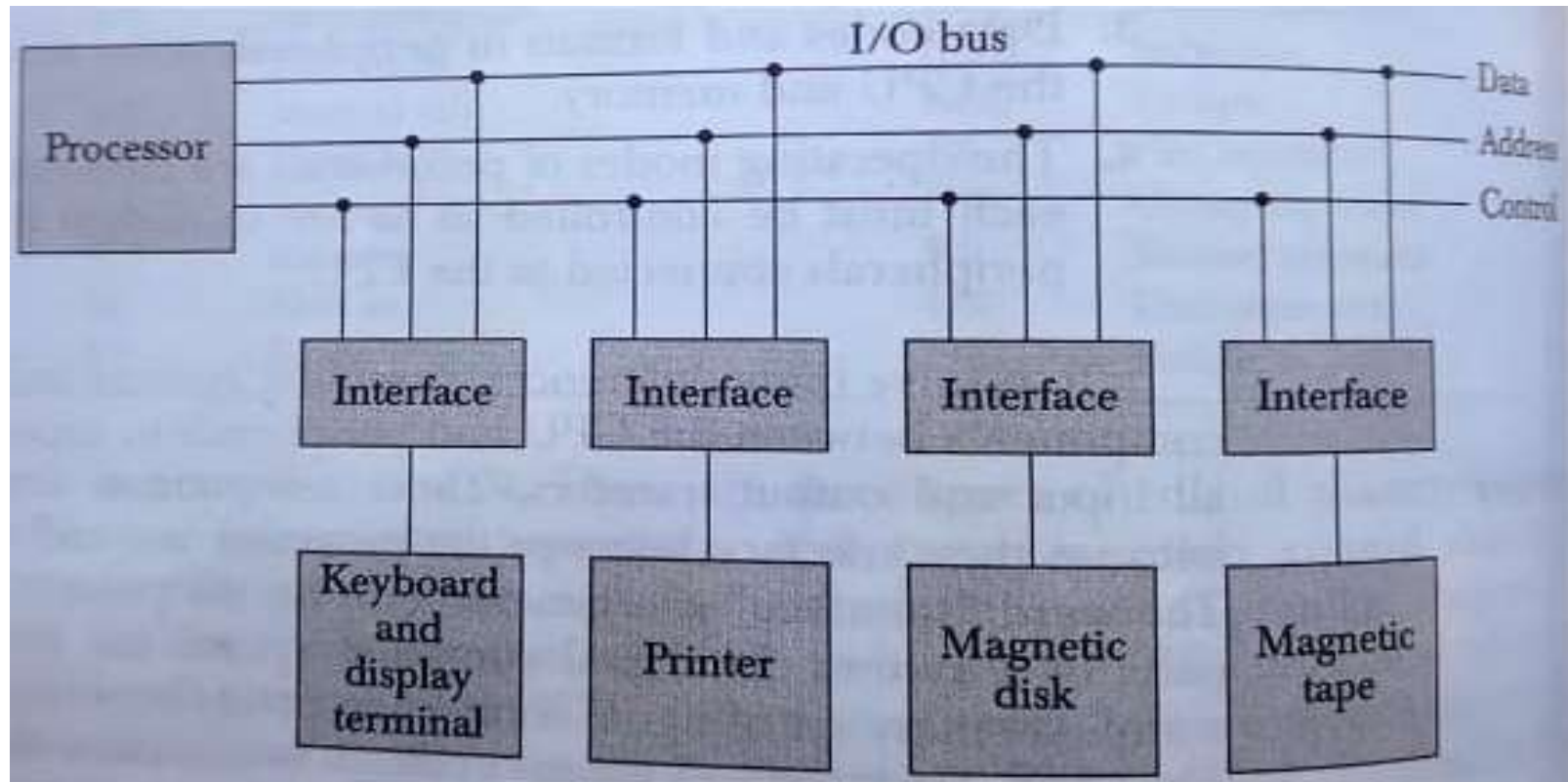
Peripheral devices:



INTERFACE

Speed and Format:

Physical orientation and signal conversion:



INTERRUPT

Signal from hardware device to get the processor attention.

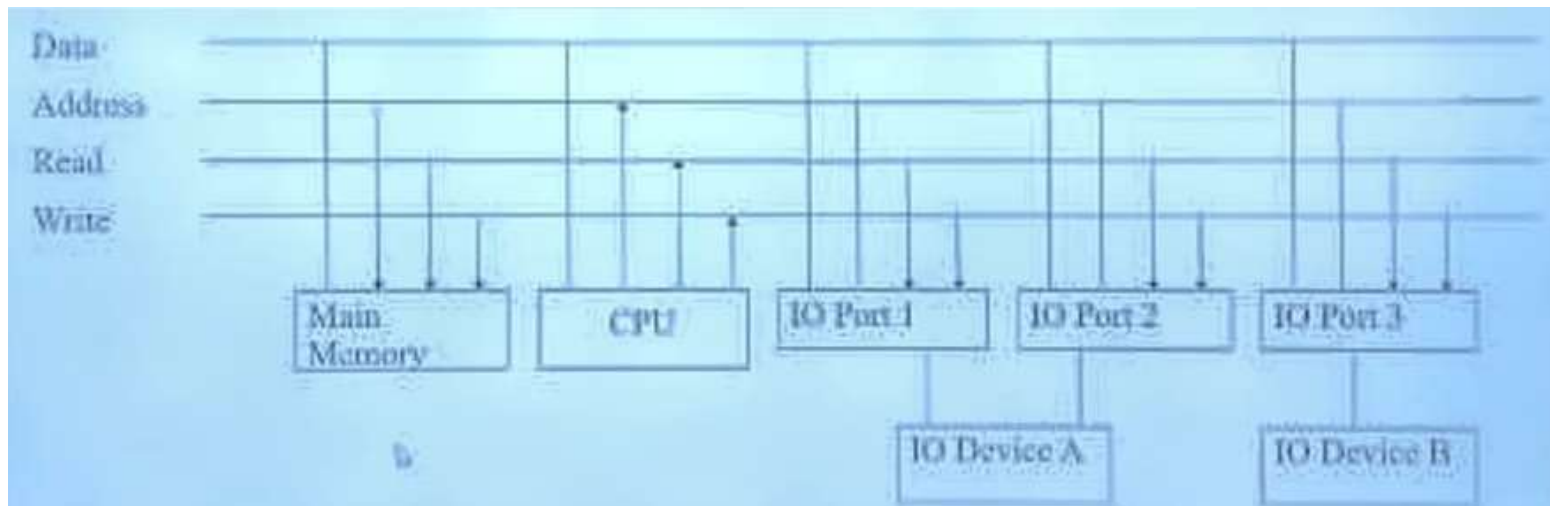
- Polled Interrupt (regular check from processor)
- Vectored Interrupt (direct ask from I/O to processor)



HOW CPU DEALS WITH MEMORY AND I/O

- Memory Mapped I/O:

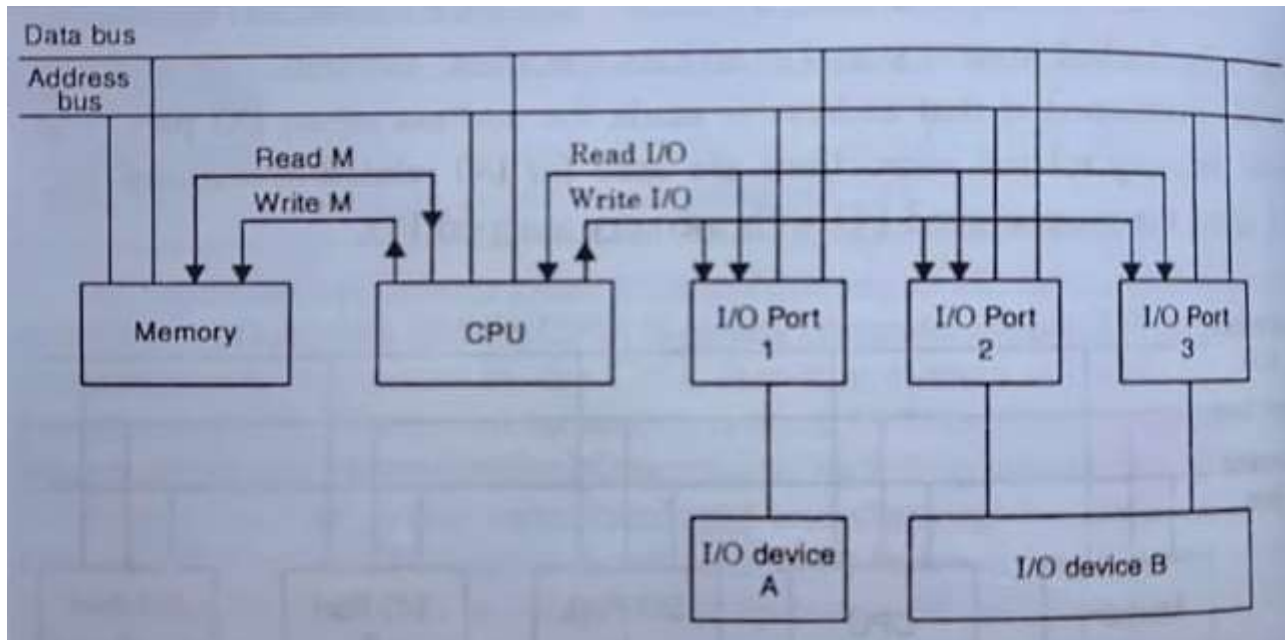
No separate I/O instructions. Here CPU can use memory type instructions for I/O data. Eg: 8085



- Isolated I/O:

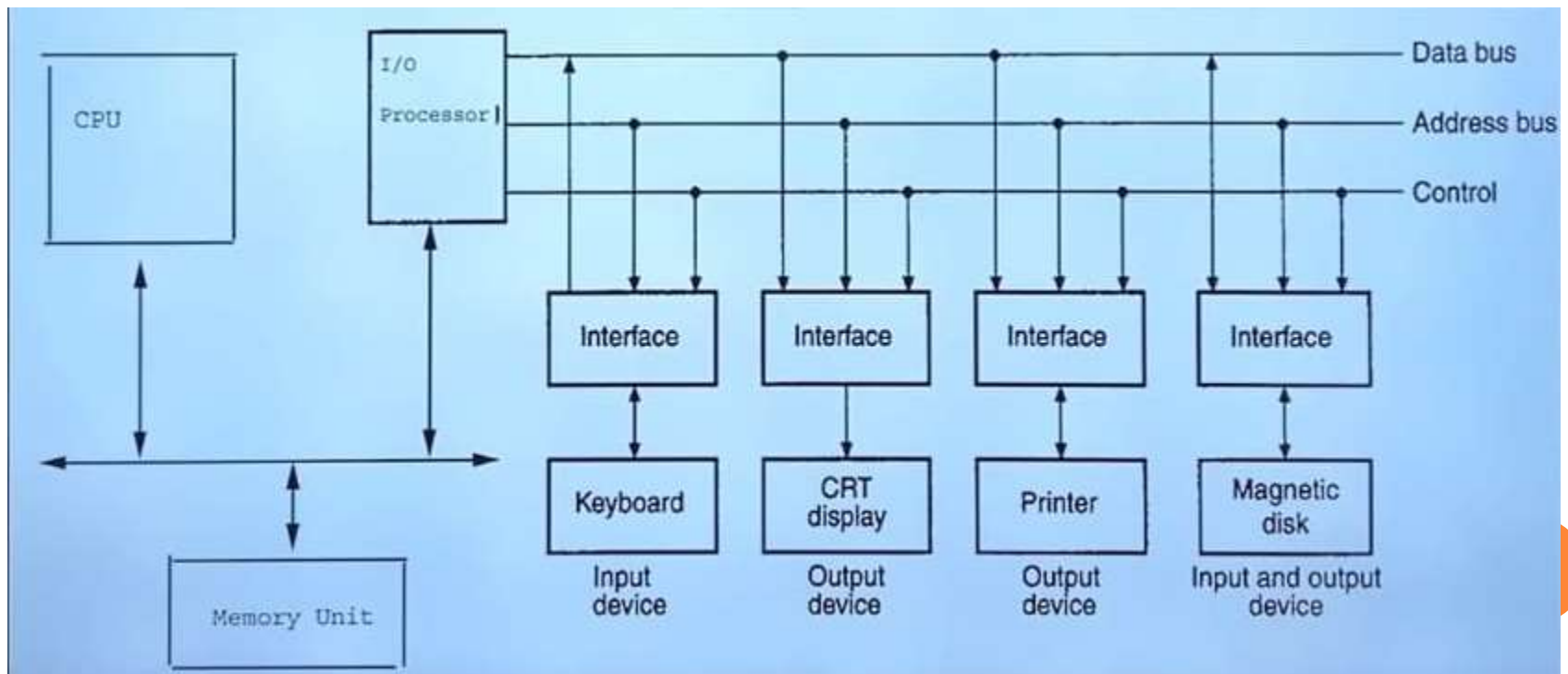
Separate read write line. I/O read and I/O write are enabled during I/O transfer, same is for memory.

Eg: 8086



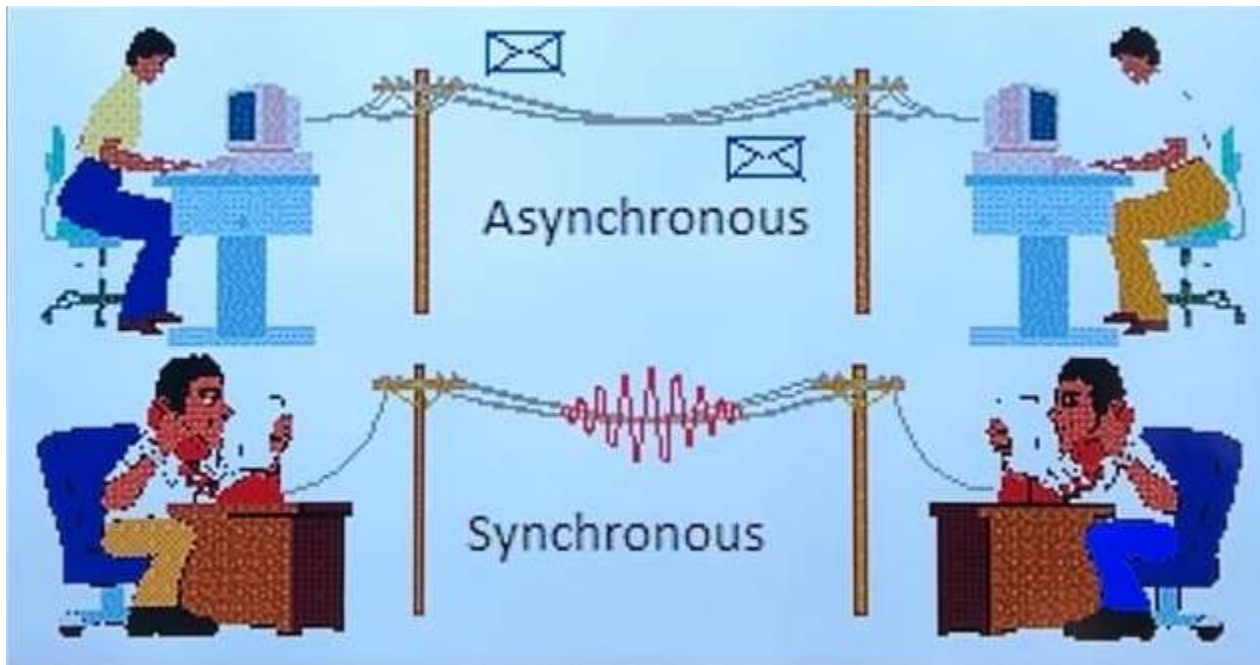
○ I/O Processor:

Independent set of address, data and control bus, one for memory and other for i/o. Separate i/o processor than CPU. Purpose of i/o proc. is to provide independent pathway for the transfer of info. between external devices and internal m/y



SYNCHRONOUS VS ASYNCHRONOUS DATA TRANSFER

- Synchronous is achieved by device called master generator (generates periodic clock).



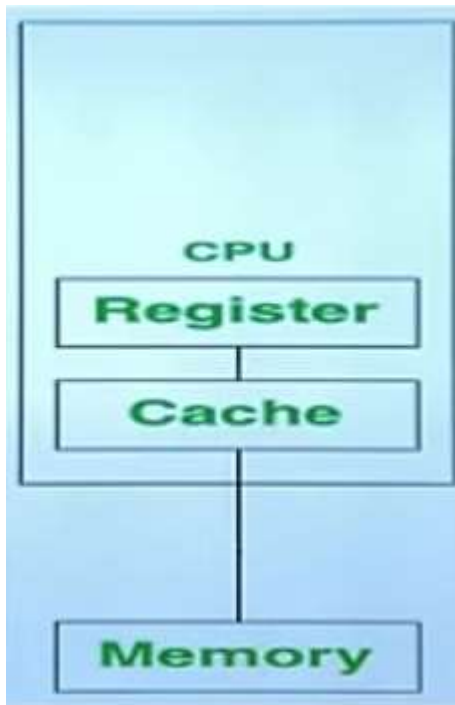
KINDS OF INTERRUPT

- Hardware Interrupt
- Software Interrupt
- Maskable Interrupt (Optional)
- Non-Maskable Interrupt (High priority demands)



PIPELINING (INTRO):

- Uni-processing Systems



Instruction	1				2			
Fetch								
Decode								
Execute								
Write								
Clock	1	2	3	4	5	6	7	8

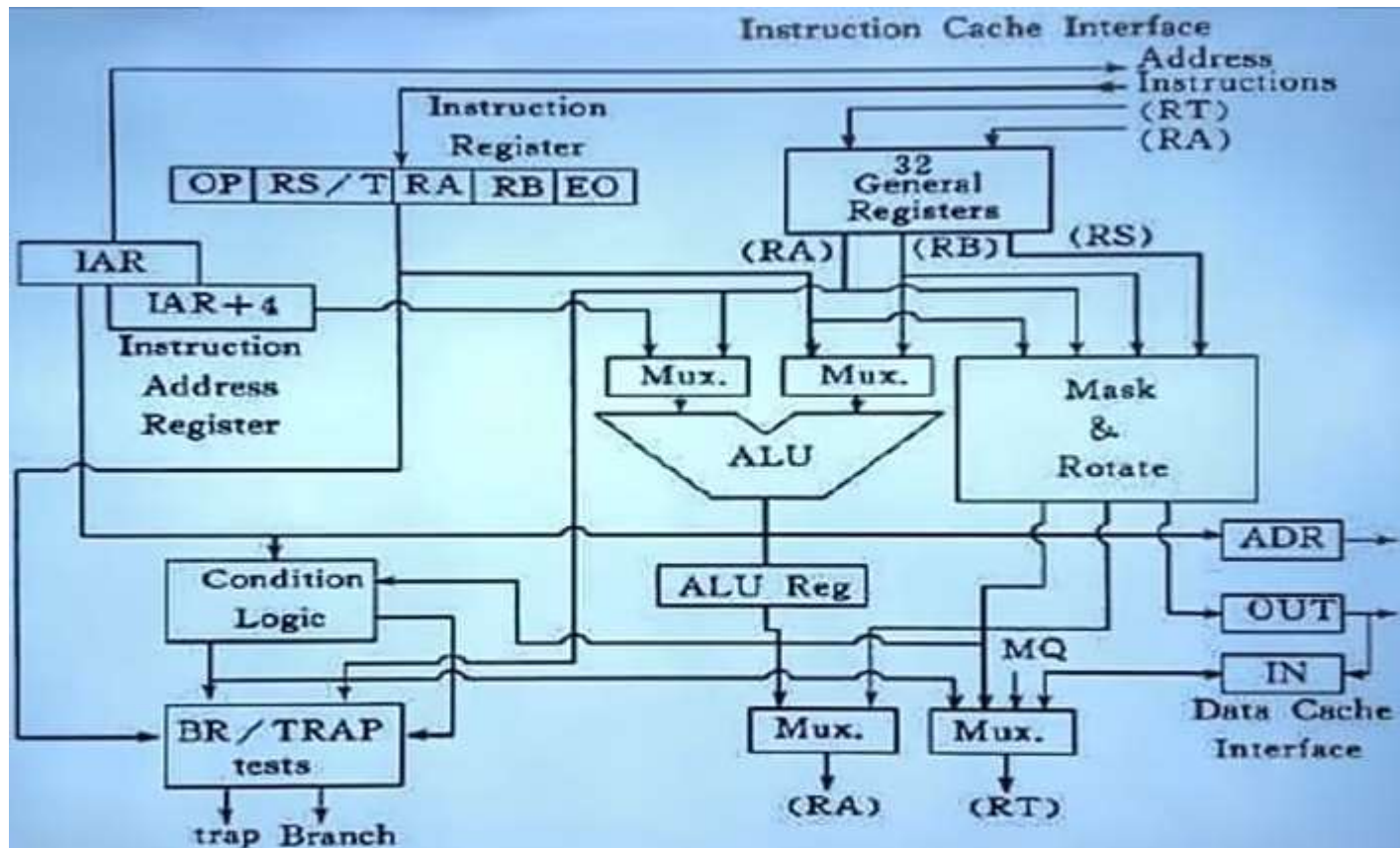
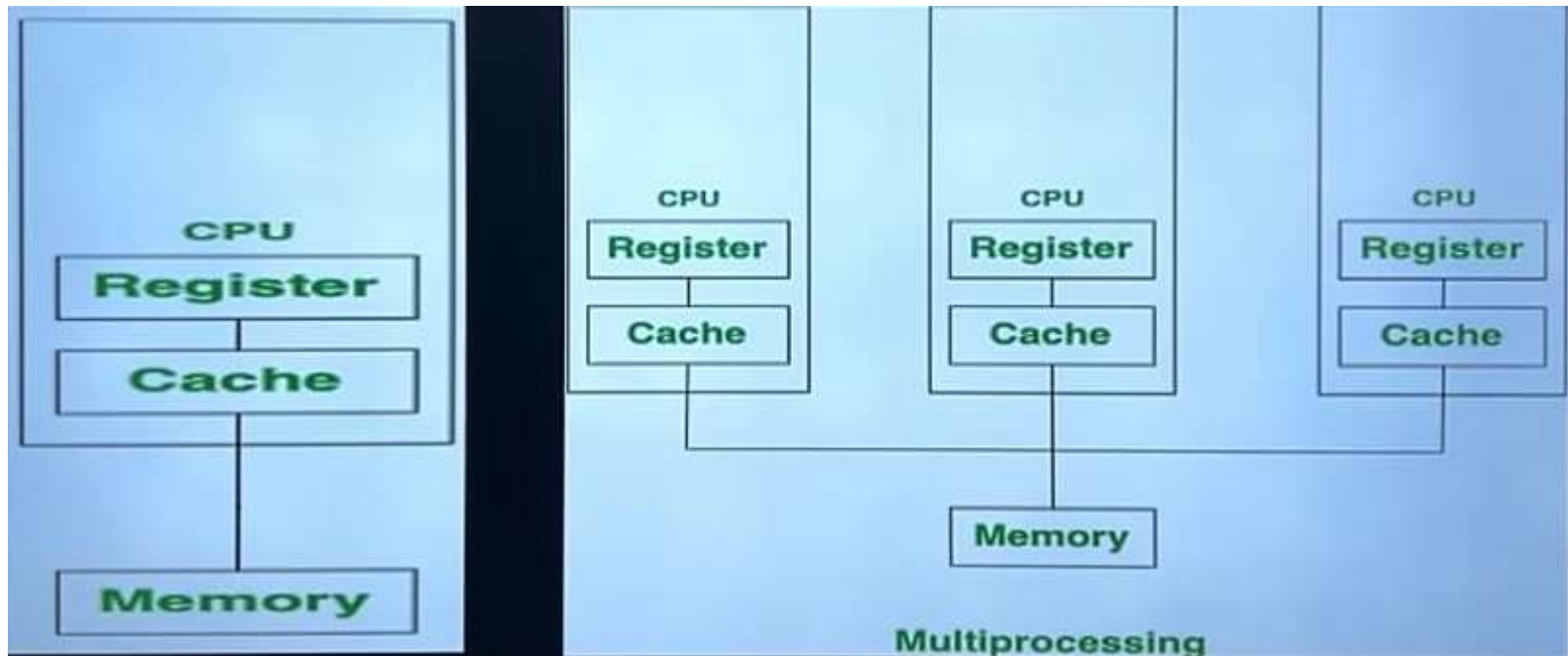


Fig. 3-2 IBM 801 Architecture

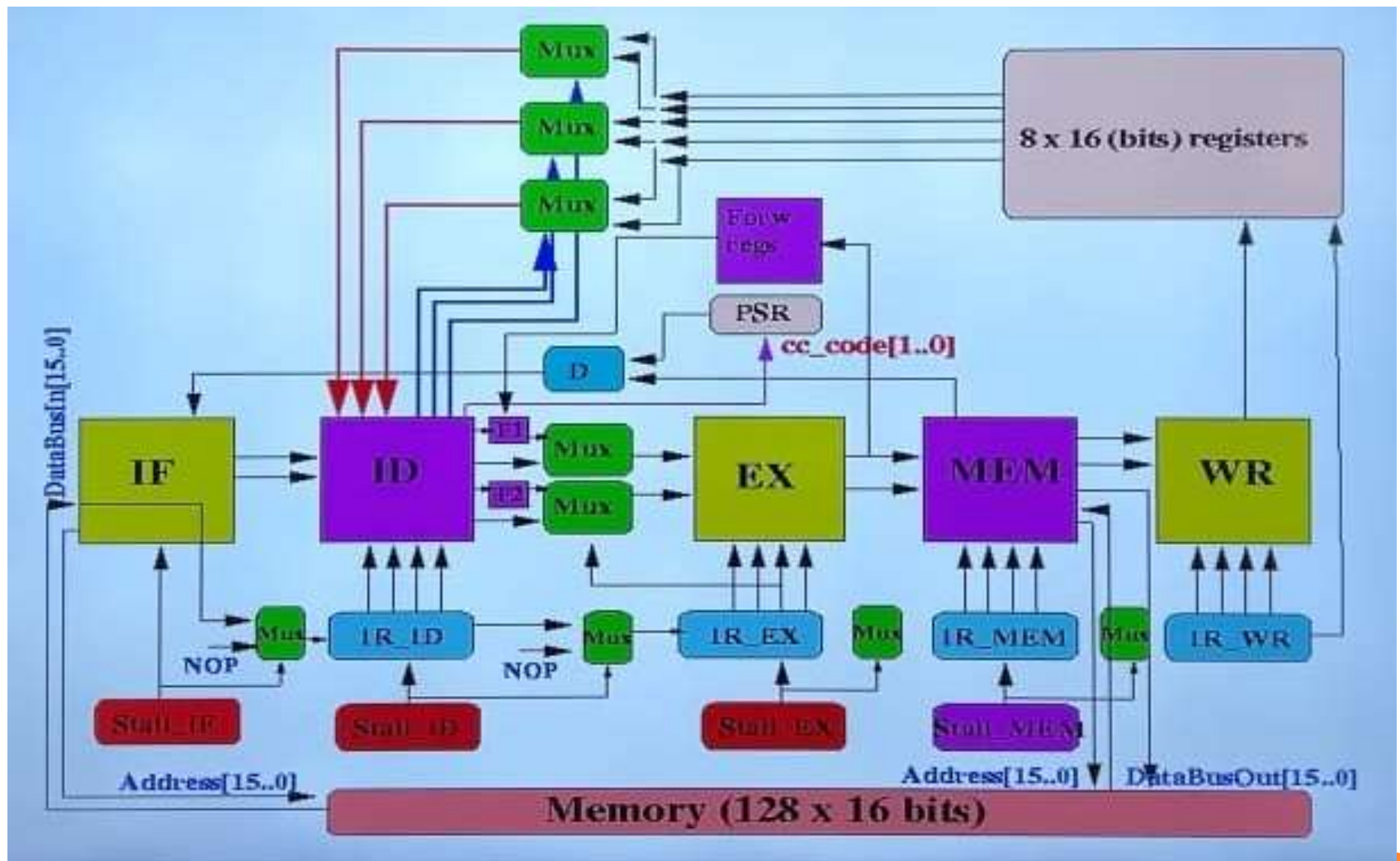
○ Uni-processing vs Multi-processing Systems



PIPELINING:

- Pipelining is a phenomenon or method using which, we will be able to run more than one instruction at the same time, on a single processor.





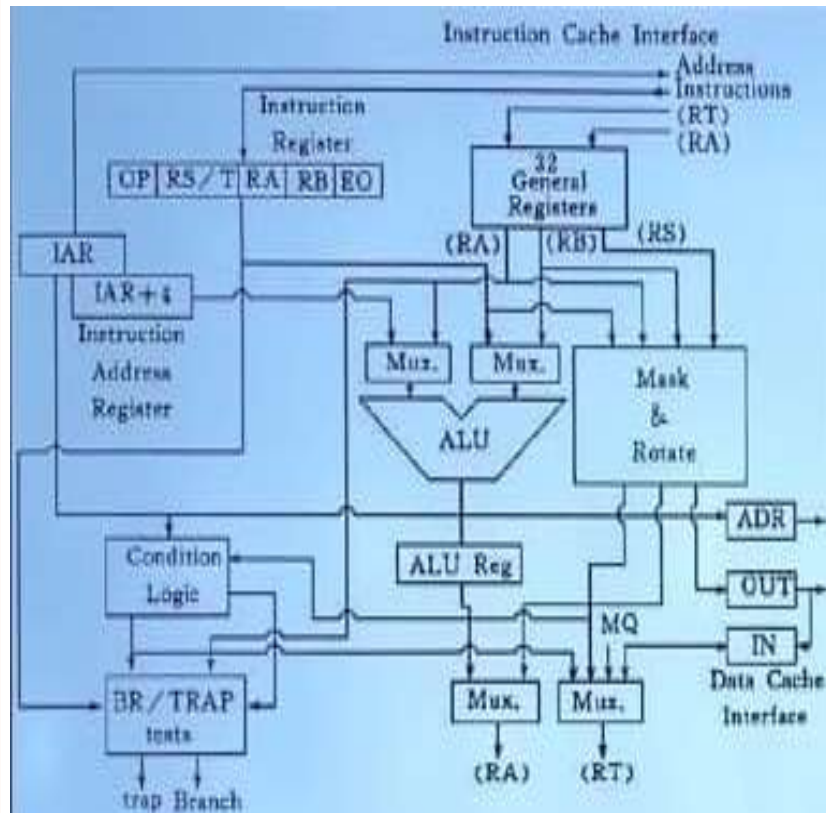
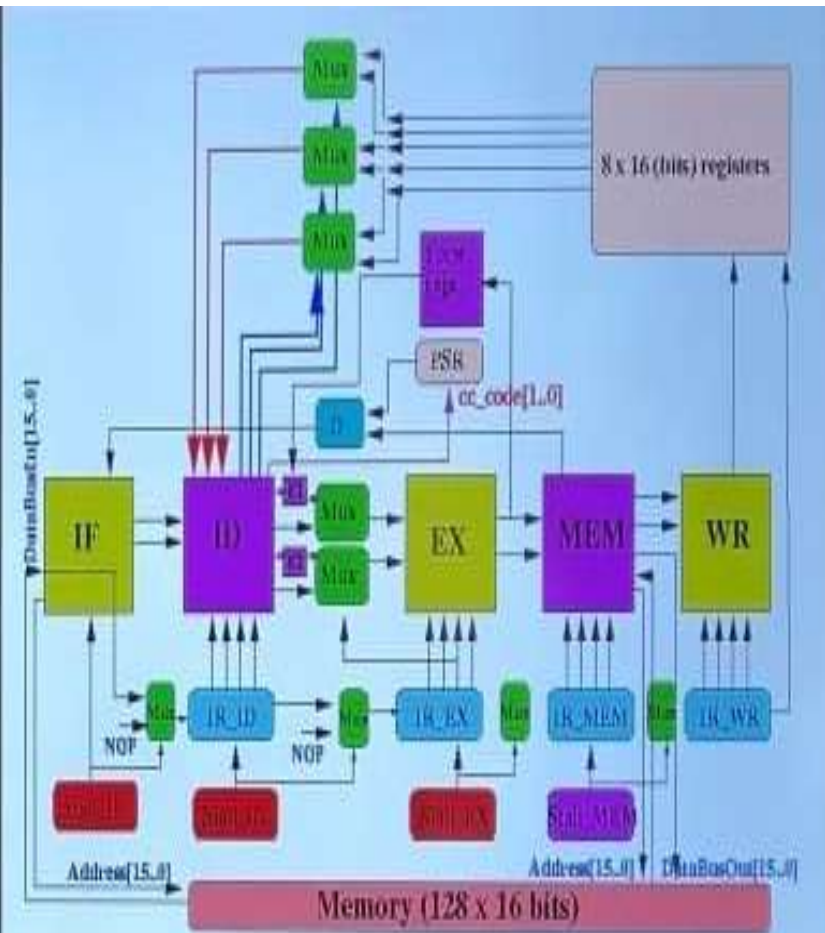
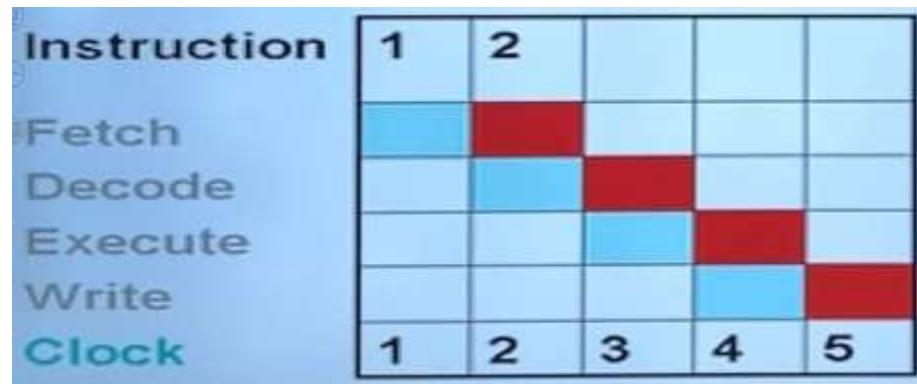


Fig. 3-2 IBM 801 Architecture



❖ Uni, Multi, and Pipelined processor working



PIPELINING IN DEPTH:

- Without Pipelining
- With pipelining
- CPI (Clock Per Instruction)
- Speedup, Efficiency



Q Consider a system where clock is triggering at a speed of 1MHz (1 clock = 1 μ s)
In a pipelined processor there are 4 stages and each stage take only 1 clock, if a program has 10 instruction then it will take what time?

- On a non-pipelined processor



	1	2	3	4	5	6	7	8	9	10	11	12	13
IF	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}			
ID		I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}		
EX			I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}	
WB				I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}

Total Time (T_p) = $[M + (N-1)] \times \text{time}$

Total Time (T_{wp}) = $M \times N$

Speedup = T_{wp} / T_p

Max Speedup = ?

Efficiency = $(\text{Speedup} / \text{Max_speedup}) \times 100$



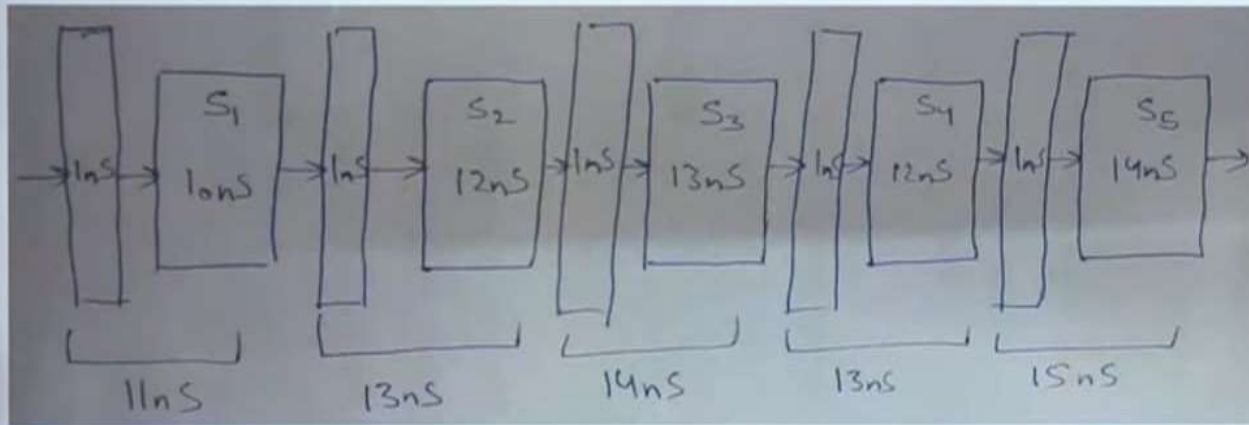
Stall:

Q Consider 5 instruction with following clock requirement?

	F	D	E	WB
I_1	1	2	1	1
I_2	1	2	2	1
I_3	2	1	3	2
I_4	1	3	2	1
I_5	1	2	1	2

[illegible]

Q After considering this diagram what must be the frequency of the processor to ensure that work of every stage will complete in 1 clock stage wise?



	S_1	S_2	S_3	S_4	S_5
1ns					
2ns					
3ns					
5ns					
15ns					



HAZARDS

- Types of Hazards

Structural Hazards, Control Hazards, Data

Hazards

